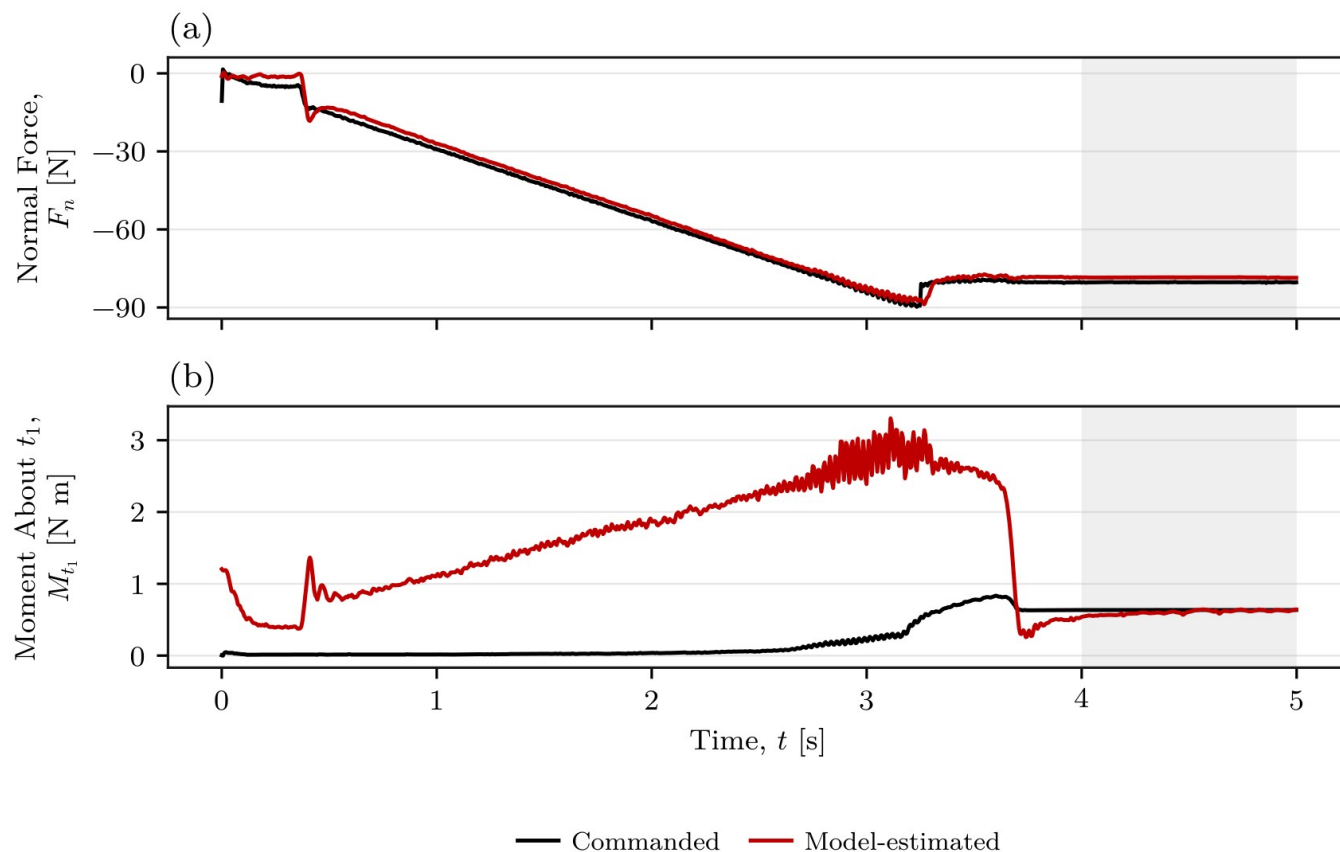


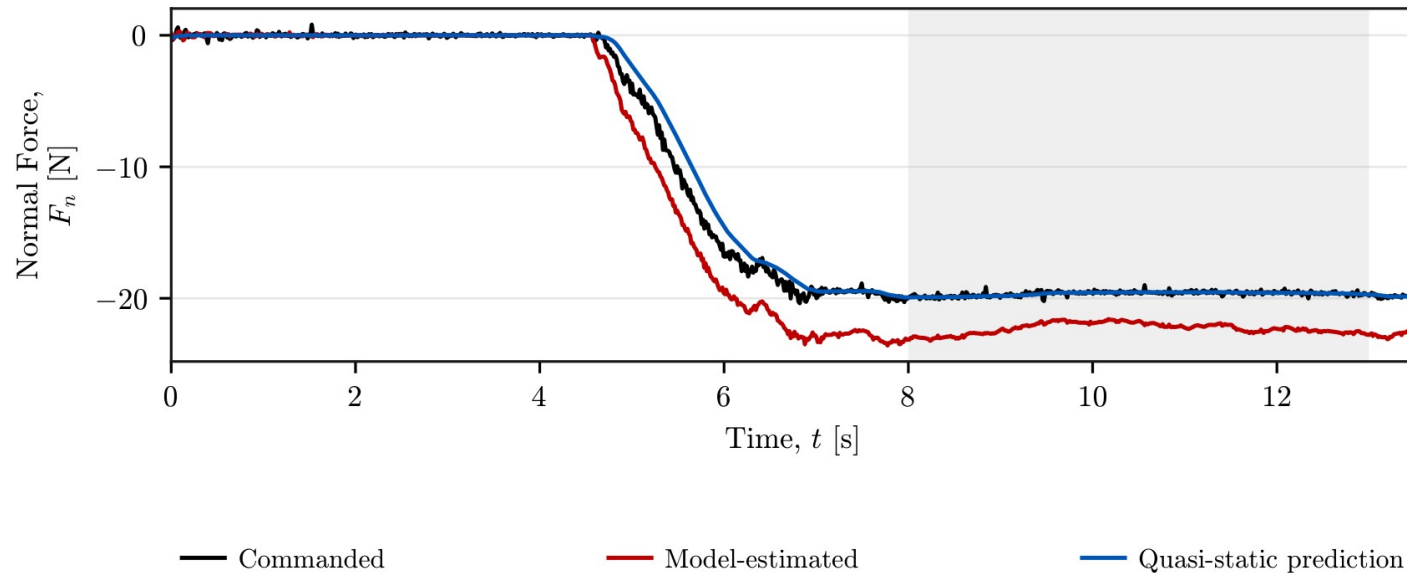
Commanded and estimated wrench



- F_n commanded normal-force component
- M_{t_1} commanded moment about the first tangent
- est model estimate from the robot
- Stationary force difference: 2.32%
- Stationary moment difference: 4.87%
- Supplementary plausibility evidence

Normal-force plausibility assessment

$$F_{n,qs} = K_{p,n} n_s^\top (e_p - e_{p,0})$$



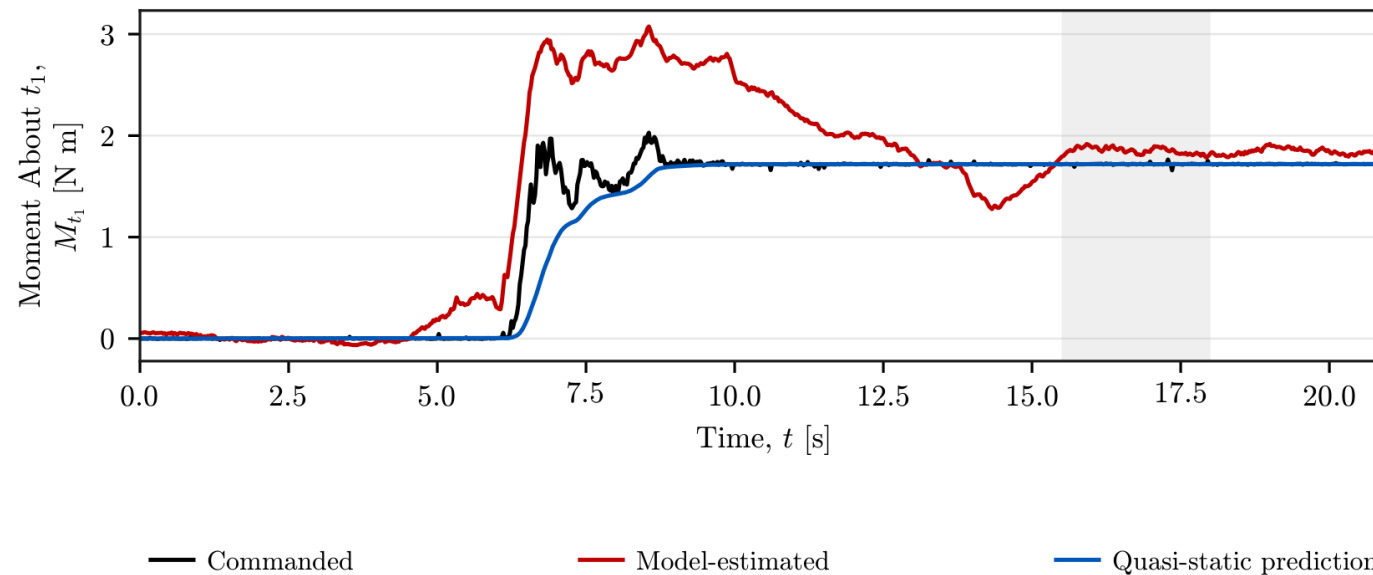
- $K_{p,n}$ normal translational stiffness
- $e_{p,0}$ position error before perturbation
- QS quasi-static spring prediction

- Spring vs command: 0.05% difference
- Estimate vs command: 13.50% difference

Moment plausibility assessment

$$M_{t_1,qs} = K_{R,t_1} t_1^\top (e_R - e_{R,0})$$

- K_{R,t_1} rotational stiffness about the first tangent
- $e_{R,0}$ orientation error before perturbation
- QS quasi-static spring prediction



- Spring and commanded mean: 1.719 N m
- Model-estimated mean: 1.852 N m